

Static Gesture Classification — Run Report

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Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Static · 35 trials
Training data path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Static · 35 trials
Training data path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Static · 35 trials
Training data path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Static · 35 trials
Training data path 5	/home/jestin/ThesisRepo/ML/NewTestData/5_Georgia/Static · 35 trials
Training data path 6	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Static · 35 trials
Training data path 7	/home/jestin/ThesisRepo/ML/NewTestData/7_Henry/Static · 35 trials
Training data path 8	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Static · 36 trials
Training data path 9	/home/jestin/ThesisRepo/ML/NewTestData/9_Joselyn/Static · 30 trials
Training data path 10	/home/jestin/ThesisRepo/ML/NewTestData/10_Josh/Static · 35 trials
Training data path 11	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Static · 36 trials
Training data path 12	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Static · 35 trials
Training data path 13	/home/jestin/ThesisRepo/ML/NewTestData/13_Nat/Static · 37 trials
Training data path 14	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Static · 35 trials
Training data path 15	/home/jestin/ThesisRepo/ML/NewTestData/16_Bella/Static · 35 trials
Total training paths	15 · 524 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Static
Report output dir	/home/jestin/ThesisRepo/ML/NewReports/StaticReports/Configuration3

Sensor selection

Hands	left, right
Segments	thumb, index, middle, ring, pinky, wrist
Modalities	YPR, Accel, Flex
Sensor columns	82 channels
Classes	7: Double_Closed_Fist, Double_Nothing, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Sensor grid (✓ enabled, ✗ disabled, — not present)

Hand / Segment	IMU_mid	IMU_prox	Flex_MCP	Flex_PIP
left palm	✗	✓	—	—
left thumb	✗	✓	✗	✓
left index	✗	✓	✗	✓
left middle	✗	✓	✗	✓
left ring	✗	✓	✗	✓
left pinky	✗	✓	✗	✓
left wrist	✗	—	—	—
right palm	✗	✓	—	—
right thumb	✗	✓	✗	✓
right index	✗	✓	✗	✓
right middle	✗	✓	✗	✓
right ring	✗	✓	✗	✓
right pinky	✗	✓	✗	✓
right wrist	✗	—	—	—

Enabled: 22 · Disabled: 24 (channels per enabled IMU depend on modality flags above).

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	minmax
Feature mode	stats+fft → feature dim 4428
Test size · seed · CV folds	0.2 · random_state=42 · 5-fold stratified
Sample counts	train (post-aug)=2095 · test=105

Augmentation (training only)

Enabled	yes
Copies per train trial	4
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.9, 1.1)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

CV = 5-fold cross-validation. Test = hold-out accuracy. M-* = macro-averaged, W-* = support-weighted. Prec / Rec / F1 on hold-out test.

Algorithm	CV acc	CV std	CV F1	CV F1 σ	Test	M-Prec	M-Rec	M-F1	W-Pre c	W-Rec	W-F1
Random Forest	0.9976	0.0030	0.9976	0.0030	0.9238	0.9327	0.9211	0.9200	0.9317	0.9238	0.9209

Per-class metrics (hold-out test)

=== Random Forest ===

	precision	recall	f1-score	support
Double_Closed_Fist	0.80	1.00	0.89	16
Double_Nothing	0.91	0.71	0.80	14
Double_Okay	1.00	1.00	1.00	15
Double_Open_Palm	1.00	1.00	1.00	15
Double_Phone	1.00	0.73	0.85	15
Double_Pistol	0.94	1.00	0.97	15
Double_Spiderman	0.88	1.00	0.94	15
accuracy			0.92	105
macro avg	0.93	0.92	0.92	105
weighted avg	0.93	0.92	0.92	105

External test (NewTestData)

Algorithm	External test acc
Random Forest	0.8571

Per-class external accuracy

Class	Random Forest	Support
Double_Closed_Fist	1.000	5
Double_Nothing	0.800	5
Double_Okay	1.000	5
Double_Open_Palm	1.000	5
Double_Phone	0.200	5
Double_Pistol	1.000	5
Double_Spiderman	1.000	5